

Artificial Intelligence

Section 3

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Section Content

- Introduction To Searching
- Types Of Search Algorithms
- Breadth-First Search
- Solving Romania Problem

- Problem-solving agents
 - In Artificial Intelligence, **Search** techniques are universal problem-solving methods.
 - **Rational agents** or **Problem-solving** agents in AI mostly used these search strategies or algorithms to solve a specific problem and provide the best result.
 - Problem-solving agents are the goal-based agents and use atomic representation. In this topic, we will learn various problem-solving search algorithms.
- Properties of Search Algorithms:
 - **Completeness**: A search algorithm is said to be complete if it guarantees to return a solution if at least any solution exists for any random input.
 - **Optimality**: If a solution found for an algorithm is guaranteed to be the best solution (lowest path cost) among all other solutions, then such a solution for is said to be an optimal solution.
 - **Time Complexity**: Time complexity is a measure of time for an algorithm to complete its task.
 - **Space Complexity**: It is the maximum storage space required at any point during the search, as the complexity of the problem.

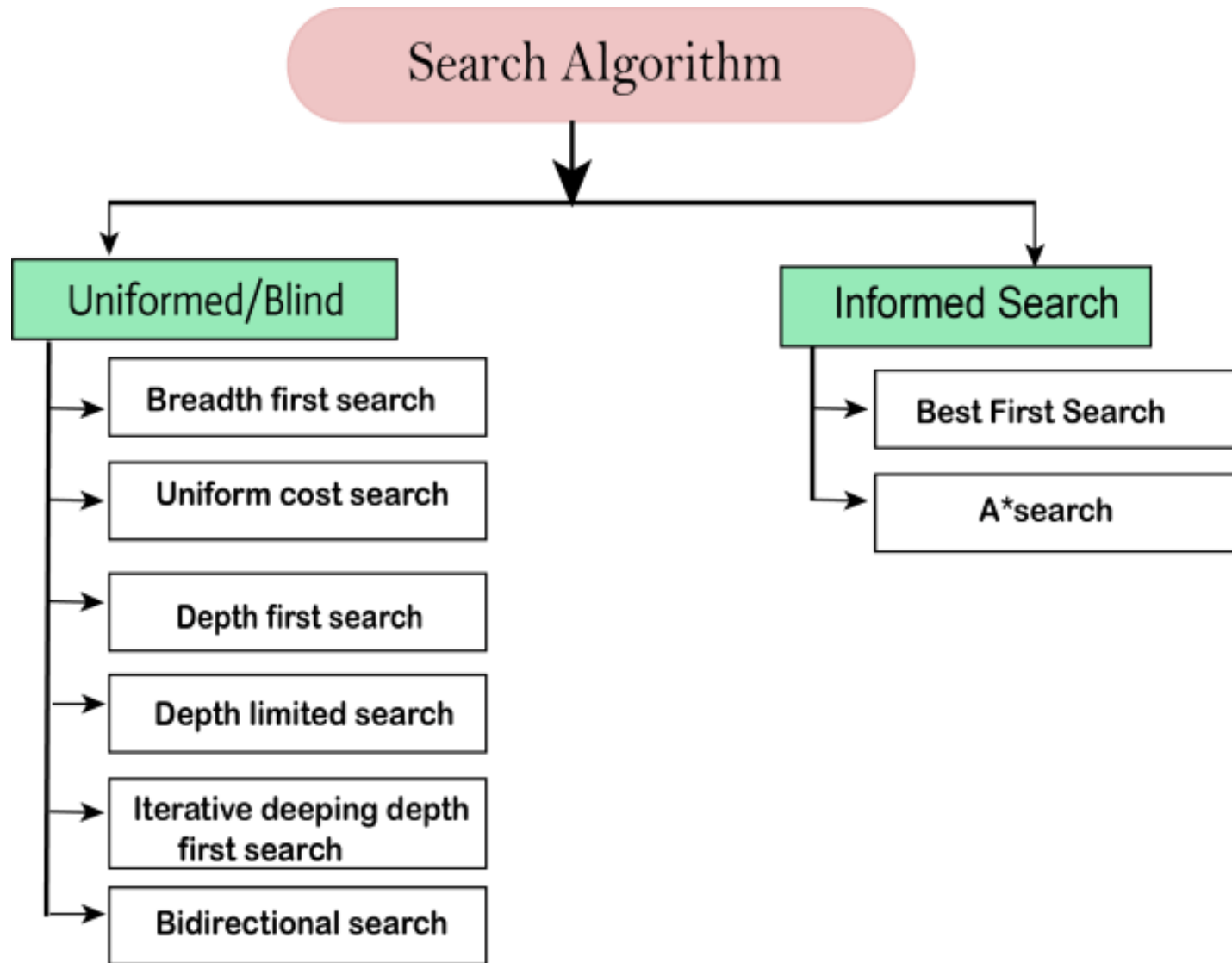
- Based on the search problems we can classify the search algorithms into **uninformed (Blind search)** search and **informed search (Heuristic search)** algorithms.
- Uninformed/Blind Search:
 - The uninformed search does not contain any domain knowledge such as closeness, the location of the goal. It operates in a brute-force way as it only includes information about how to traverse the tree and how to identify leaf and goal nodes. so, it is also called blind search. It examines each node of the tree until it achieves the goal node.
- Informed Search
 - Informed search algorithms use domain knowledge. In an informed search, problem information is available which can guide the search. Informed search strategies can find a solution more efficiently than an uninformed search strategy. Informed search is also called a Heuristic search.
 - A heuristic is a way which might not always be guaranteed for best solutions but guaranteed to find a good solution in reasonable time.

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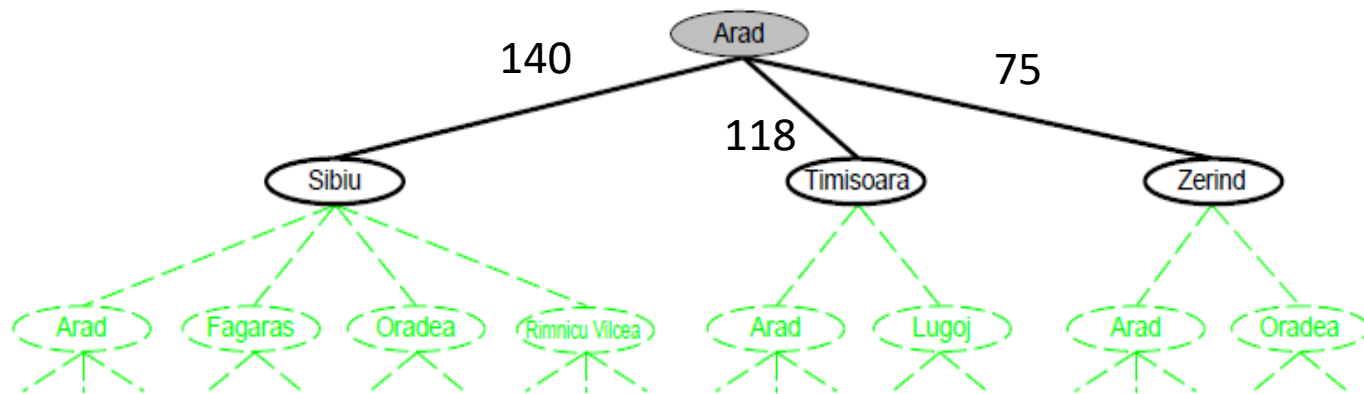
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Basic tree search

```
function TREE-SEARCH(problem, strategy) returns a solution, or failure
  initialize the search tree using the initial state of problem
  loop do
    if there are no candidates for expansion then return failure
    choose a leaf node for expansion according to strategy
    if the node contains a goal state then return the solution
    else expand the node and add the resulting nodes to the search tree
  end
```



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- Definition

- Breadth-first search is the most common search strategy for traversing a tree or graph. This algorithm searches breadthwise in a tree or graph, so it is called breadth-first search.
- BFS algorithm starts searching from the root node of the tree and expands all successor node at the current level before moving to nodes of next level.

- Advantages

- BFS will provide a solution if any solution exists.
- If there are more than one solutions for a given problem, then BFS will provide the minimal solution **which requires the least number of steps.**

- Disadvantages

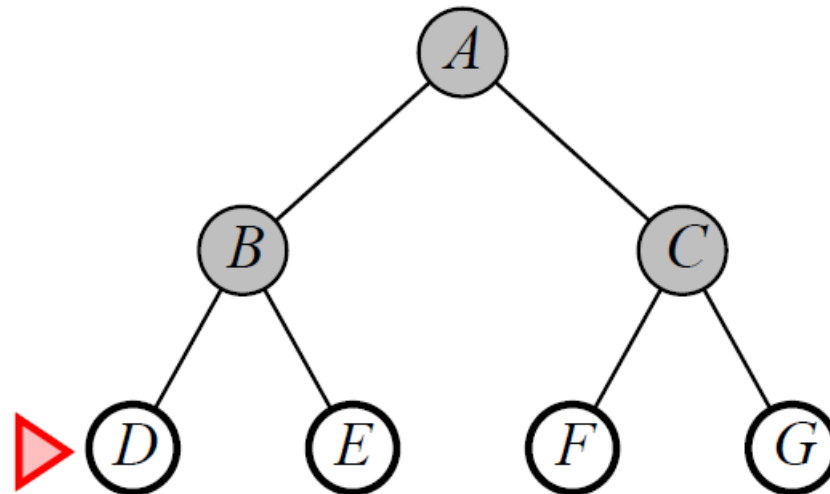
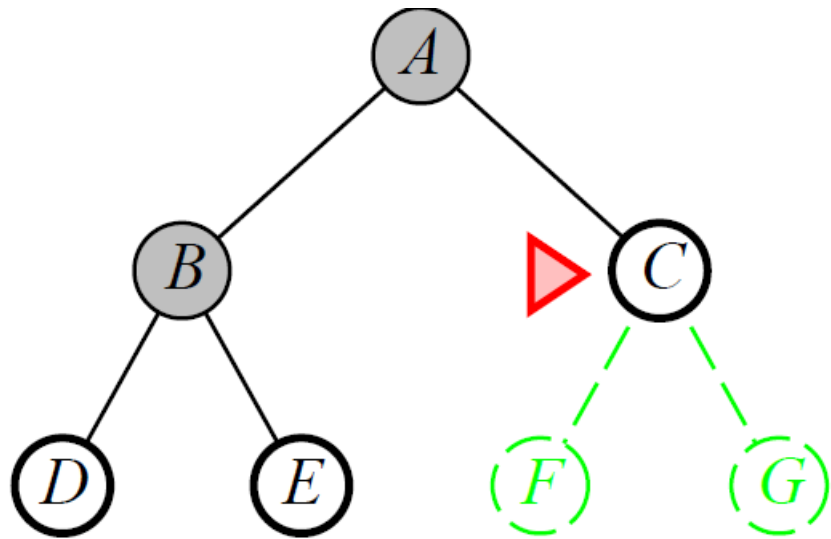
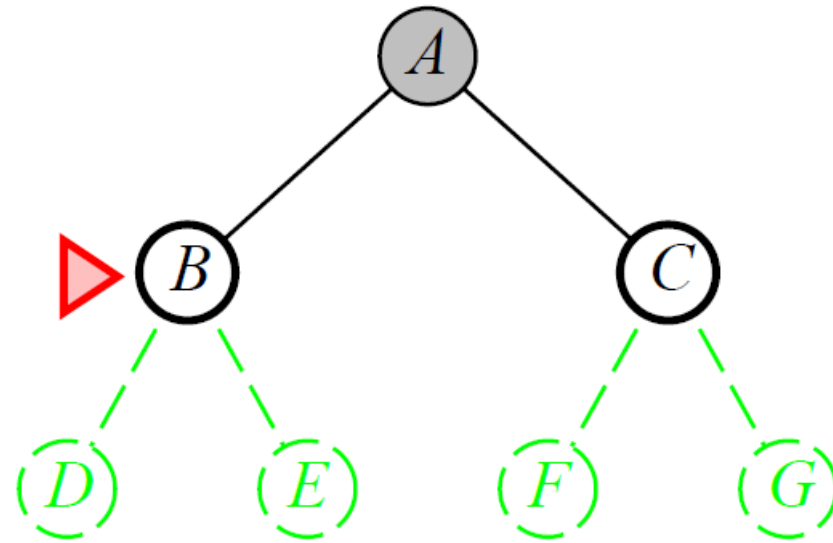
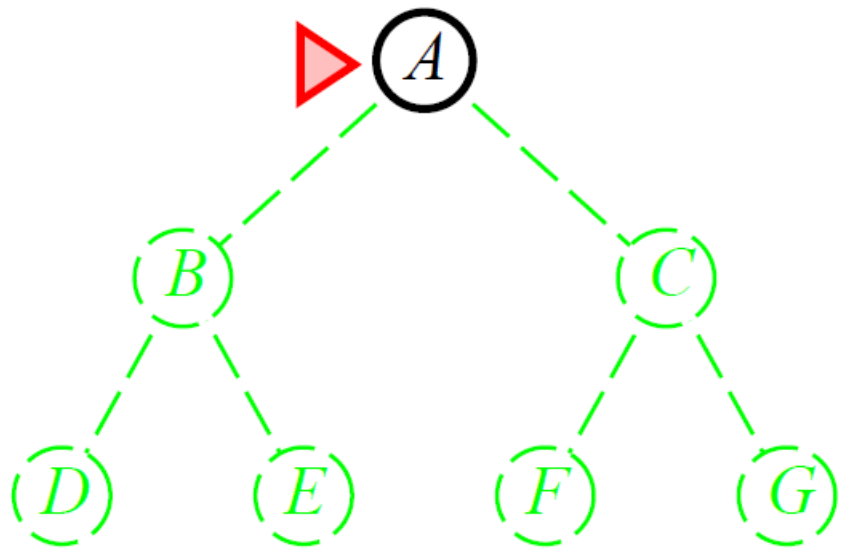
- It requires lots of memory since each level of the tree must be saved into memory to expand the next level.
- BFS needs lots of time if the solution is far away from the root node.

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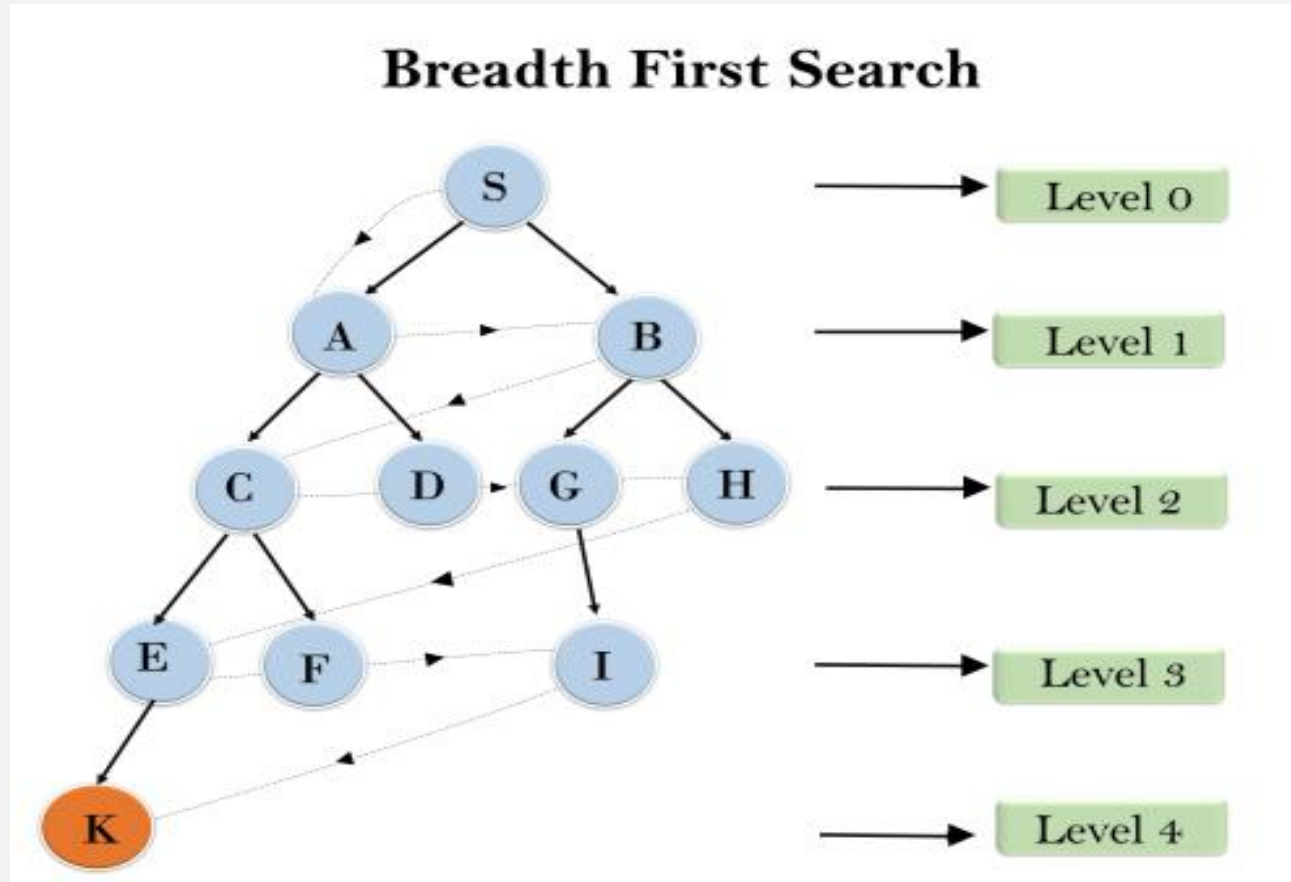
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- In the below tree structure, we have shown the traversing of the tree using BFS algorithm from the root node S to goal node K. BFS search algorithm traverse in layers, so it will follow the path which is shown by the dotted arrow, and the traversed path will be:



S----> A---->B---->C---->D---->G---->H---->E---->F---->I---->K

- Breadth-first search implemented using FIFO **queue** data structure.

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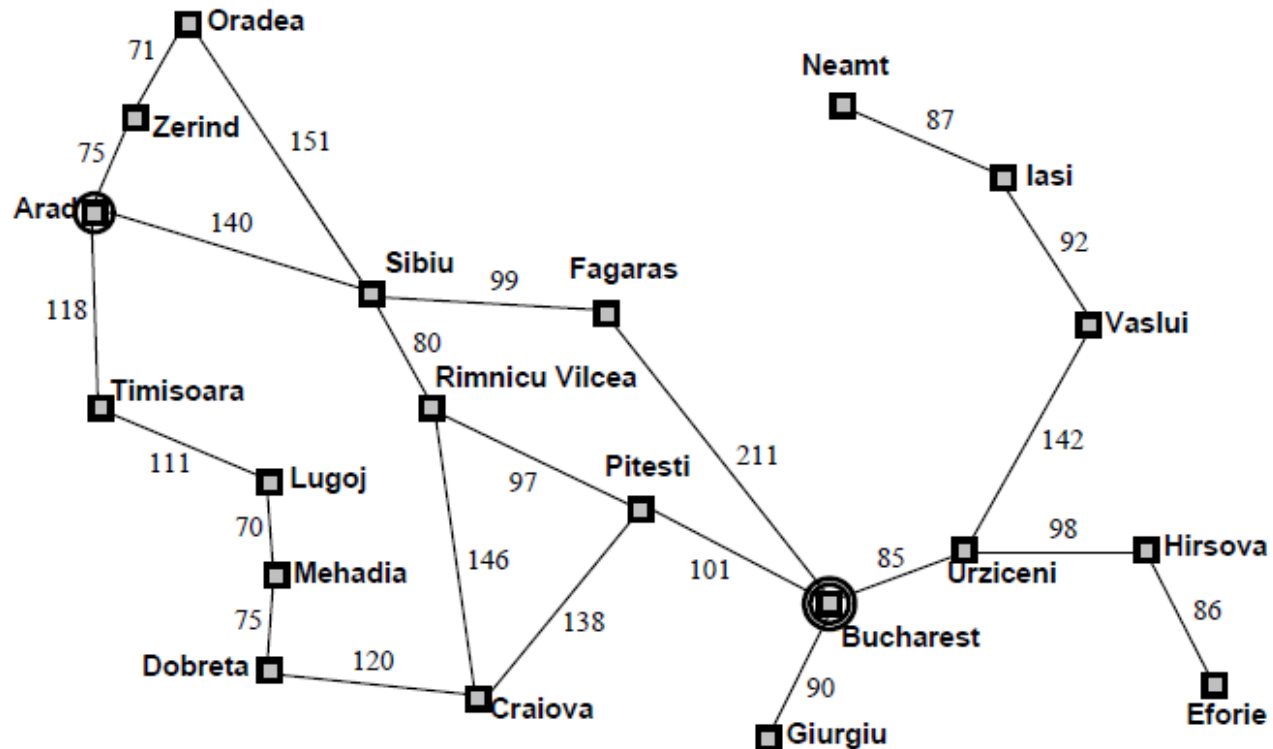
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- **Problem formulation** is the process of deciding what actions and states to consider, given a goal.
- **Goal formulation** based on the current situation and the performance measure, is the first step in problem solving.
- We assume that the environment is **observable** , so the agent always knows the current state. We also assume the environment is **discrete**, so at any given state there are only finitely many actions to choose from. We will assume the environment is **known**, so the agent knows which states are reached by each action. Finally, we assume that the environment is **deterministic**, so each action has exactly one outcome.
- Under these assumptions, the solution to any problem is a fixed sequence of actions.

Romania Problem

- **Problem:** On holiday in Romania; currently in Arad. Flight leaves tomorrow from Bucharest. Find a short route to drive to Bucharest.



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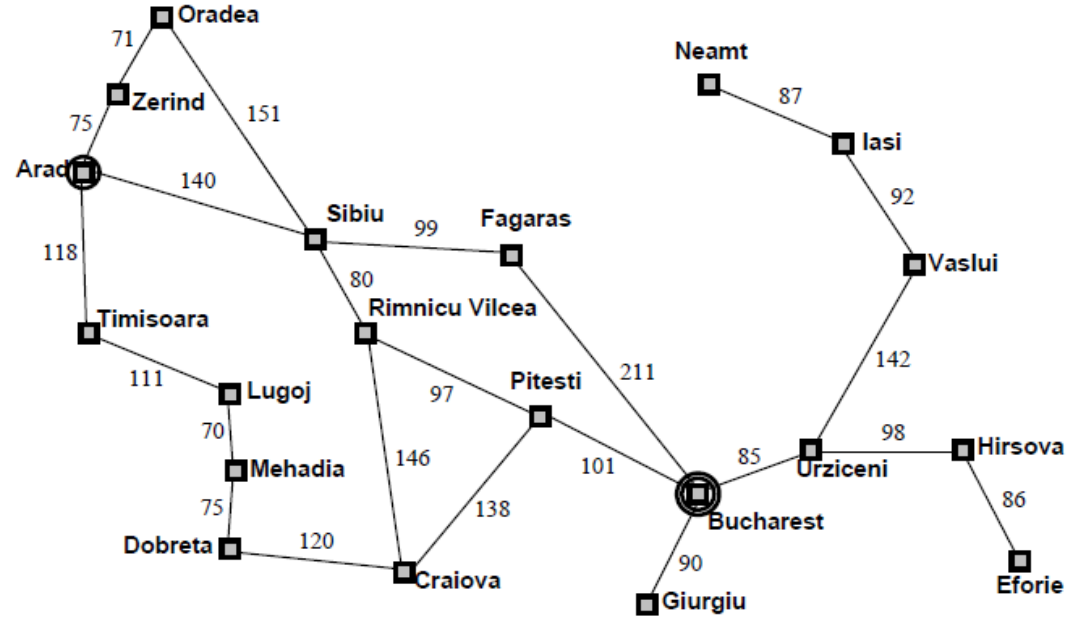
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- Search Algorithm Terminologies:
 - **Search:** Searching is a step-by-step procedure to solve a search-problem in a given search space. A search problem can have three main factors:
 - **Search Space:** Search space represents a set of possible solutions, which a system may have.
 - **Start State:** It is a state from where agent begins the search.
 - **Goal test:** It is a function which observe the current state and returns whether the goal state is achieved or not.
 - **Search tree:** A tree representation of search problem is called Search tree. The root of the search tree is the root node which is corresponding to the initial state.
 - **Actions:** It gives the description of all the available actions to the agent.
 - **Transition model:** A description of what each action do, can be represented as a transition model.
 - **Path Cost:** It is a function which assigns a numeric cost to each path.
 - **Solution:** It is an action sequence which leads from the start node to the goal node.
 - **Optimal Solution:** If a solution has the lowest cost among all solutions.

Romania Problem



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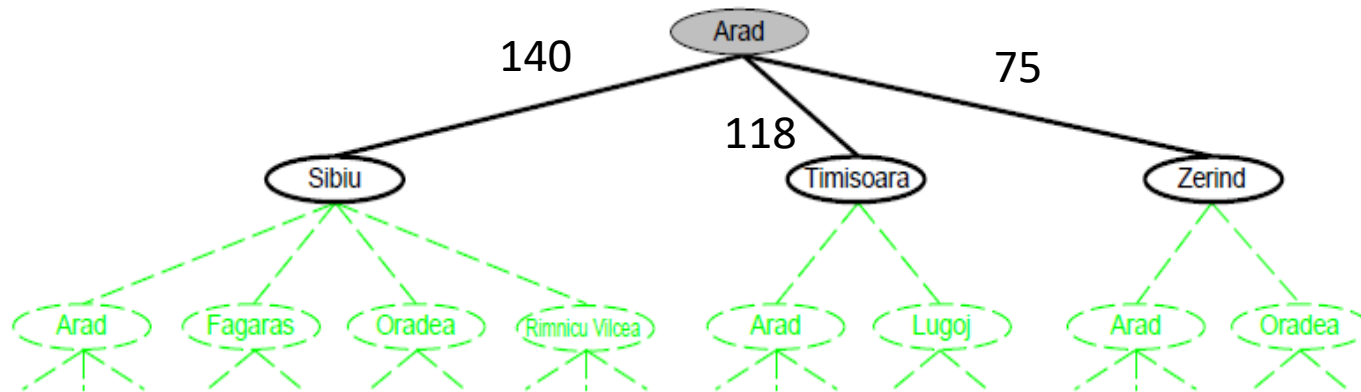
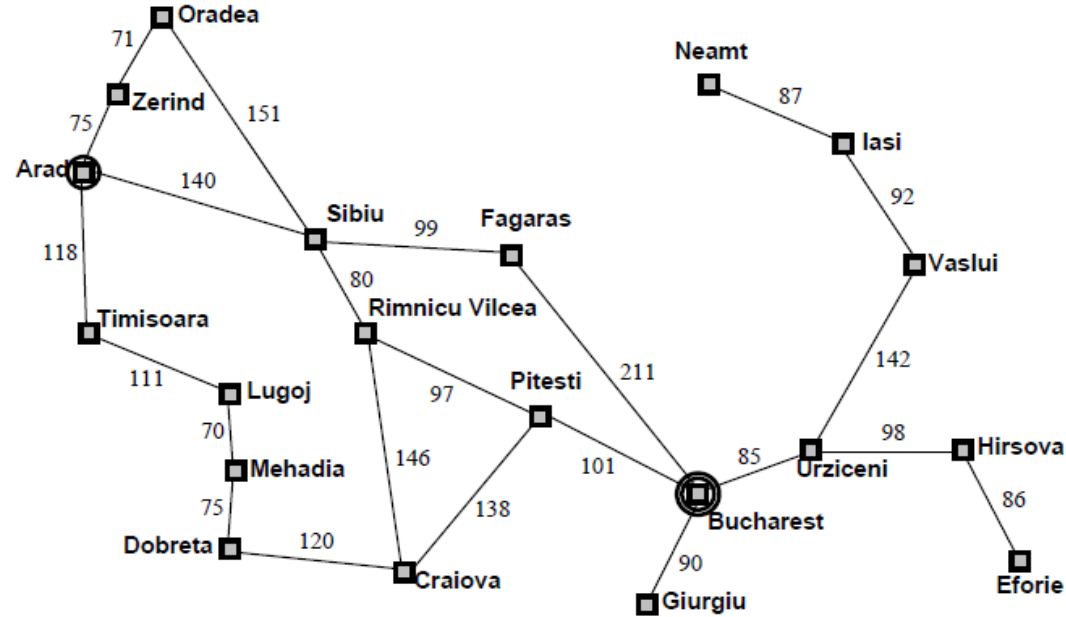
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- **Search Space:** various cities
- **Start State:** Arad
- **Goal test:** Is the current city is Bucharest
- **Actions:** drive between cities
- **Transition model:** Action(Arad) = {Zerind, Sibiu, Timisoara} , ...
- **Path Cost:** (Arad, Zerind) = 75, (Arad, Zerind, Oradea) = 146, ...
- **Solution:** {Arad, Zerind, Oradea, Sibiu, Fagaras, Bucharest}, {Arad, Sibiu, Fagaras, Bucharest}, ...
- **Optimal Solution:** {'Arad', 'Sibiu', 'Rimnicu Vilcea', 'Pitesti' , 'Bucharest'}

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Breadth-first Search Pseudocode

function BREADTH-FIRST-SEARCH(*problem*) **returns** a solution, or failure

node $\leftarrow$ a node with STATE = *problem*.INITIAL-STATE, PATH-COST = 0

if *problem*.GOAL-TEST(*node*.STATE) **then return** SOLUTION(*node*)

frontier $\leftarrow$ a FIFO queue with *node* as the only element

explored $\leftarrow$ an empty set

loop do

if EMPTY?(*frontier*) **then return** failure

node $\leftarrow$ POP(*frontier*) /* chooses the shallowest node in *frontier* */

 add *node*.STATE to *explored*

for each *action* **in** *problem*.ACTIONS(*node*.STATE) **do**

child $\leftarrow$ CHILD-NODE(*problem*, *node*, *action*)

if *child*.STATE is not in *explored* or *frontier* **then**

if *problem*.GOAL-TEST(*child*.STATE) **then return** SOLUTION(*child*)

frontier $\leftarrow$ INSERT(*child*, *frontier*)

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